Homework 5: Trading and Financing a Strategy

The following sections contain exercises related to how trading is funded and how transaction costs affect the performance of trading strategies. The following questions regarding trade execution should be answered independently of each other and relate to the following limit order book:

Top of the limit order book		
	Shares	Price
Asks ---->	1900	34.56
	1700	34.54
	1200	34.53
	400	34.52
	300	34.51
Bids <----	1000	34.49
	1100	34.48
	1400	34.47
	1500	34.46
	2200	34.45

1. Bid-ask spread
What is the posted bid-ask spread in cents and basis points of the mid-price? If you buy 100 shares with a market order and immediately (i.e., before any new orders arrive or any existing orders are cancelled) sell them with a market order, then what is your P&L?

2. Limit order
A limit order to buy 150 shares at $34.50 arrives in the market. What happens? I.e., which transactions occur and what is the resulting bid-ask spread in cents? (In the rest of the exercise, assume that this order is subsequently cancelled.)

3. Walking the book
A limit buy order for 2000 shares at $34.53 arrives. Document all transactions, compute the volume-weighted average execution price of the transaction, and determine the bid-ask spread after the transaction(s).

4. Market impact curve
Compute the average execution price for market orders to buy, respectively, 10, 1000, 2000, 3000, and 4000 shares. Plot the market impact curve, that is, the execution price as a function of the size of the order. Discuss the relative importance of the bid-ask spread vs. the slope of the market impact curve for small and large traders. Also, discuss what market liquidity risk means in terms of what can happen to the limit order book in the future.

5. Transaction costs (not related to the previous exercises, only theoretical)
If you hold a stock whose t-cost is high, it is likely that your future t-costs will be high as you will have to sell the stock eventually. This implies that accounting for the current t-cost only is not enough. How should the optimal portfolio problem be modified to account for the future t-costs?

1. Bid ask spread = difference between best bid and best ask.
   Best bid = highest bid price = $34.49
   Best ask/offer = lowest offer price = $34.51
   Mid-price = ($34.49 + $34.51) / 2 = $34.5

Bid-Ask Spread in cents (difference between best ask and best bid) = ($34.51 - $34.49)*100 = 2 cents

Spread in basis points of the mid-price = (0.02 / 34.5) * 10000 = 5.8 bps

Buy $100 shares with market order and sell them immediately:

P&L = $34.49 *100 - $34.51 * 100 = -$2

2. **Limit order**

A limit order is a trade order with an instruction to trade at the best price available, if the price is better or equal to the limit price. As the lowest ask is $34.51, the limit order at $34.50 will not be executed as it is below the ask price. Therefore, there is no transaction cost occur.

3. **Walking the book**

Limit order to buy 2000 shares at 34.53

Sequence	Price	Shares	Shares Bought	Remaining
First	$34.51	300	300	= 2000 -300 = 1700
Second	$34.52	400	400	=1700 - 400 = 1300
Third	$34.53	1200	1200	= 1300 - 1200 = 100

After executing the third time at $34.53, there are 100 shares left and those shares will not be executed as the next ask price is $34.54, which is above the limit order.

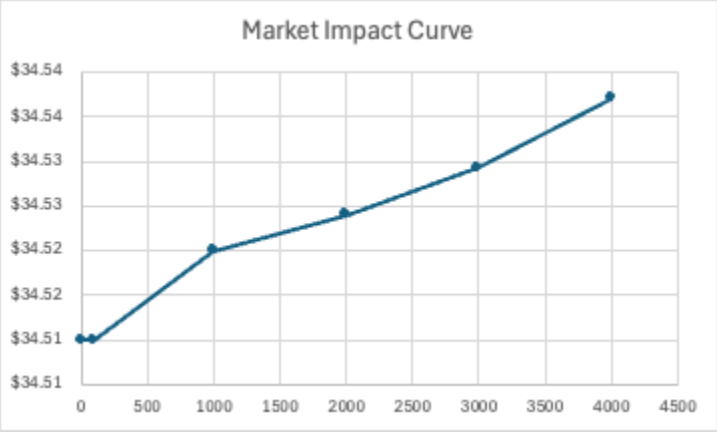
Volume weighted average execution price
= [(300 * $34.51) + (400*$34.52) + (1200 * $34.53)] / 1900 = $34.525

Bid-ask spread after the transaction:
After the transaction, the ask orders at $34.51, $34.52, and $34.52 are cleared. We have shares at $34.54 and $34.56 left.
So we have the new best ask of $34.54 and the best bid is $34.49.
The new spread after the transaction is $34.54 - $34.49 = 0.005.

4. **Market impact curve.**

Size	Average Execution Price	Market Impact (average trading - mid price)/(mid price)
10	$34.51	=(34.51-34.5) / 34.5=0.00029
100	$34.51	=(34.51-34.5) / 34.5=0.00029
1000	((300*34.51)+(400*34.52)+(300*34.53)) / 1000 = $34.520	=34.520-34.5/34.5=0.00058
2000	((300*$34.5)+(400*$34.52) + (1200*$34.53) + (100*$34.54))/2000 = $34.524	=34.524-34.5/34.5=0.00070
3000	((300*34.5)+(400*34.52)+(1200*34.53) +(1100*34.54)) /3000 = $34.529	=34.529-34.5/34.5=0.00085
4000	((300*34.5)+(400*34.52) + (1200*34.53) + (1100*34.54) + (1000*34.56)) /4000 = $34.537	=34.537-34.5/34.5=0.00107

For small trades, the impact of trade size to price does not change much. However, the slopes steeper for larger order size, showing the increasing cost for large trades, resulting in high trading costs.



Market Impact vs Bid-Ask Spread:

The bid-ask spread represents the minimum cost of trading, which measures the two best prices to buy and sell. Small orders face mainly the bid-ask spread cost.

Market impact is the additional cost that incurs when we gradually buy/sell larger amounts of shares, which also heavily affects transaction costs as we need to pay higher to execute large amounts of shares. In conclusion, large orders are influenced by both bid-bask spread cost and the market impacts cost that increase when trade volume increases.

Market Liquidity Risk

> Market liquidity risk is the risk that an investor will be unable to transact the desired quantity of an asset quickly without significantly affecting its price.
>
> The risk will depend on the size of the order book. The risk will lead to higher prices for even small trades, and lead to steeper market impact curves for shallow order books.

If liquidity worsens in the future

Bid-ask spread will get larger. Market impact curve will be steepen, which leads to higher price for larger orders. This will increase transaction costs, especially for large and urgent trades.

5. Transaction costs

Trading costs or transaction costs are all the costs that the trader incurs when transacting on the market and which are deducted from the sum of money they spend when transacting.

To adjust the optimal portfolio:

- We can either reduce the expected return by incorporating transaction costs when calculating expected returns. This adjustment decreases the net expected return of each asset and reflects the true profitability after accounting for transaction costs.
- Or we also can include transaction costs directly in the portfolio optimization model to account for the cost of trading when determining the optimal portfolio weights.